Human-Computer Interaction and Digital Accessibility



Organization – Team Leadership

- **Giulio Berti**
- **Lorenzo Felix**



Organization – CSS Leaders

- **Emile Bui**
- **Maria Miteva**



Organization – The Git Leader

- **Edoardo Beretta**



Topics – Input/Output devices

Team members: **Vlad-Mihai Lungu** (leader), **Francesco Corti**, **Modou Lamin Camara**, **Yifei Wang**

Draft sub-topics:

- History of Input Devices
  - Evolution from punch cards and keyboards to mice and touchscreens.
- Evolution of Visual Outputs
  - Development of displays from CRT to LCD, OLED, and Retina.
- Haptic & Audio Feedback
  - Evolution of vibration, force feedback, and audio interaction.
- Alternative & Niche I/O Devices
  - Specialized peripherals for gaming, industry, and research.



Topics – Future of human/machine Interfaces

Team members: **Edoardo Fiorini** (leader) **Jonah De Lieme**, **Seo Dazio**

Draft sub-topics:

- Brain-Computer Interfaces (BCI)
  - Brain-controlled devices and their medical applications.
- Spatial Computing & VR/AR
  - Virtual and augmented reality, hand tracking, and 3D interaction.
- Assistive Technologies for Healthcare
  - Advanced prosthetics and adaptive interfaces for disabilities.
- Ambient Computing & Voice UI
  - Smart environments controlled through voice and gestures.



Topics – Designing human/machine interfaces (information system engineering)

Team members: **Mirco Togni** (leader) **Alessandro Campanari**, **Massimiliano Cipriani**

Draft sub-topics:



Usability & UX Principles	Principles for creating intuitive and user-friendly interfaces.
Cognitive Ergonomics & Laws of UX	Human perception, mental models, Fitts's Law, and Hick's Law.
Information Architecture	Organizing content, navigation, and information hierarchies.
Mobile & Responsive UI Design	Adapting interfaces for mobile, desktop, and wearable devices.

Topics – Web accessibility

Team members: **Shalini Denti** (leader) **Manuel Pollazzon**, **Andrii Synelnyk**

Draft sub-topics:

WCAG Guidelines & Standards

Inclusive Design Principles

Economics of Accessibility

Assistive Web Tools

Web accessibility principles and compliance levels.

Designing interfaces for users with diverse abilities.

Financial, SEO, and legal benefits of accessible design.

Screen readers, magnifiers, and keyboard navigation.

Topics – design failures

Team members: **Tommaso Russo** (leader) **Emile Bui**, **Daniel Lula**, **Boukhezna Mohamed Chaker**

Draft sub-topics:

Historic UI Design Failures	Major interface failures and their consequences.
Dark Patterns & Deceptive Design	Manipulative interfaces that influence user decisions.
Ethical HCI & Attention Economy	Digital addiction, persuasive design, and user autonomy.
Bad Mobile UX & Misleading Touch Targets	Small buttons, intrusive pop-ups, and poor mobile layouts.



GIT documentation

The GitHub repository offers the following methods of documenting for the moment:

- CONTIBUTING.md
- README.md
- /docs/contributing/git-guide.pdf



CONTRIBUTING.md

Contributing.md is a file demonstrating how to contribute, explaining in basics branching and pull requests.

This file serves as a tutorial on the needed commands to contribute to the project.

Some examples on the right:

Type	When	Example
add	new files or new content	*add: home page*
edit	changes to something that already exists	*edit: menu colors*
fix	something was broken and now works	*fix: menu on mobile*
rm	removing files or content	*rm: old footer images*
docs	README, CONTRIBUTING, the guide	*docs: add commit rules*

README.md

The readme.md file represents the most basic manual for the project; it is the first step to take to contribute and understand the project.





Git Guide

- It serves as a more elaborated report on the functioning of the versioning system we will use for this project.
- It contains more in-depth information on **pull**, **branching**, commands and general rules and best practices to follow.
- The file also acts as a **general FAQ** for possible issues and problems.



Communication Channels

General Group: For project-wide discussions, announcements, and communication between teams.

CSS Design Team: To coordinate CSS development, maintain design consistency, and provide technical assistance.

Git Repository Team: To manage repository organization, discuss issues, pull requests, merges, and overall project structure.

Topic Leaders Team: To coordinate with topic leaders, discuss content development, and oversee the organization and implementation of HTML pages.

Project Structure

```
project/
├── README.md
├── source/
│   ├── html/
│   │   ├── topic 1/
│   │   │   ├── subtopic-1.html
│   │   │   ├── subtopic-2.html
│   │   │   └── ...
│   │   ├── ...
│   │   └── topic N/
│   │       └── ...
│   ├── css/
│   │   ├── topic 1/
│   │   └── style.css
│   └── images/
│       └── ...
└── docs/
    └── contributing/
        ├── CONTRIBUTING.md
        ├── git-guide.pdf
        └── git-guide.tex
```



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Thank you for your attention

